

1(a)	(named) buffer ;	1	
1(b)	ref. to first mixing the algae with (sodium) alginate ; ref. to then adding (alginate and algae) to <u>calcium</u> chloride ; ref. to method of dropping mixture (to form beads) ;	3	e.g. using syringe or pipette
1(c)(i)	<i>independent variable:</i> light intensity / distance of container from lamp ; <i>dependent variable:</i> colour of indicator / pH ;	2	

Question	Answer	Marks	Guidance
1(c)(ii)	<i>five from:</i> 1 method to vary light intensity ; 2 five values of stated distances in the range 10 to 200 cm ; 3 cover tubes with metal foil / light proof cover ; <i>critical step</i> 4 expose to light and record, colour change / pH, after fixed time ; 5 ref. to a suitable control ; 6 same, (stated) volume / concentration, of hydrogencarbonate solution ; 7 same, number / mass / volume, of algal balls ; 8 ref. to a minimum of two replicates <u>and</u> calculate a mean <i>or</i> identify / eliminate / remove anomalies ; 9 medium risk or hazard <u>and</u> suitable safety precaution ;	5	<i>ignore 'amount' for any volume or concentration</i> 1. e.g. distances between tube and lamp or different strengths of neutral density filters or wattages of bulbs / variable resistor / dimmer switch / number of lamps 2 if any other method used, must refer to 5 stated values e.g. wattages / etc. 3 A work in a dark room with only one light source 4 A any stated time A time to a specific colour / pH 5 e.g. hydrogencarbonate with, glass beads / balls with no algae / balls with dead algae 9 e.g. sodium hydrogencarbonate indicator solution is harmful / irritant and wear gloves / goggles / mask A allergy to, alginate / algae, and wear gloves / mask / goggles

Question	Answer	Marks	Guidance
1(c)(iii)	place a heat filter between the lamp and container ;	1	A description of heat filter e.g. container of water / double glazed glass sheet A cold light source e.g. LED lights
1(d)	1.6×10^{-3} / 0.0016 / 0.002 ;	1	
1(e)(i)	<i>one from:</i> judging colour alone is subjective ; different people see colours differently ; <i>idea that</i> pH probe is more sensitive ;	1	A gives a numerical / quantitative result
1(e)(ii)	curve should be to the right of original curve and must be drawn from 0 to 10 minutes ;	1	
1(e)(iii)	indicator solution provides, hydrogencarbonate ions / carbon dioxide ;	1	
1(f)(i)	there is no <u>significant</u> difference between the (mean) pH for treatment 1 and treatment 2 / no weed killer and weed killer A ; or the difference between the (mean) pH for treatment 1 and treatment 2 / no weed killer and weed killer A is not <u>significant</u>	1	
1(f)(ii)	<i>one from:</i> continuous data ; data is normally distributed ; standard deviations are approximately the same ; size of the two samples is less than 30 ;	1	I 'compare the means'
1(f)(iii)	n = 10 ;	1	

Question	Answer	Marks	Guidance
1(f)(iv)	yes for weed killers A and C / no for weed killer B ; <i>plus two from:</i> (significant for A or C) because p value / probability, is less than 0.05 or p value shows there is less than 5% chance that results are due to chance ; effect of weed killer A greater than weed killer C (because less than 0.001) ; (not significant for B) because p value greater than (0).05 or p value shows there is less than 5% chance that results are due to weed killer B ;	3	